

Total No. of Questions : 8]

SEAT No. :

PD-4486

[Total No. of Pages : 2

[6403]-294

T.E. (E&TC)

Honors in Robotics

ROBOT PROGRAMMING & SIMULATION

(2019 Pattern) (Semester - VI) (304183HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume Suitable data if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Explain motion control, hand control, program control commands used in robotic programming with example [6]

b) Briefly explain the Production rate calculations used for robot in detail [6]

c) Explain any three basic commands in VAL-II with example. [6]

OR

Q2) a) W.r.t. VAL-II programming language explain simple pick and place application. [6]

b) Explain WAIT, SIGNAL and DELAY commands used in robotics for communications using simple application. [6]

c) With schematic diagram, explain the robotic applications in welding industry. [6]

Q3) a) Explain manual and automatic mode of operation of industrial robot. [6]

b) With simple example explain move master command language. [6]

c) With suitable example explain different constants and variables used in AML Language of robot. [6]

P.T.O.

OR

- Q4)** a) Describe Pick and place operation using Industrial robot. [6]
b) With suitable example explain different Program control statements are used in AML language of robot. [6]
c) Describe the different Motion & Sensor commands of AML language of robot. [6]
- Q5)** a) Write short note on Virtual robotics. [5]
b) Discuss the concept of soft robotics? Explain Robotic Process Automation in details. [6]
c) Discuss in detail about Robot studio online software. [6]

OR

- Q6)** a) What is role of Computer vision in AR and VR. [6]
b) Discuss machine interference w. r.t. Multiple robots. [5]
c) Write advantages and disadvantages of simulation. [6]
- Q7)** a) Explain Distributed lag models in details. [5]
b) What are the basics of simulation? Discuss different Steps in simulation [6]
c) Describe role of simulation in decision making. [6]

OR

- Q8)** a) Discuss different techniques of simulation? Explain monte carlo method in details. [7]
b) Compare simulation packages with programming languages. [5]
c) Describe general purpose simulation packages in detail. [5]

